Computational Fluid Dynamics - 15/6/2022

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We want to simulate the natural convection flow in a square cavity induced by the imposition of different temperatures on the lateral sides of the cavity. The flow is governed by the following steady incompressible Navier–Stokes equations with the *Boussinesq buoyancy model* on the domain $\Omega = [0,1] \times [0,1]$:

$$\begin{cases} -\nu \triangle \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p + \beta \mathbf{g}(T - T_{\text{ref}}) = 0, & \text{in} \quad \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in} \quad \Omega, \\ -k \triangle T + \mathbf{u} \cdot \nabla T = 0, & \text{in} \quad \Omega, \end{cases} \tag{1}$$

where $\mathbf{u}$, p and T are velocity, pressure and temperature, respectively, ν is the kinematic viscosity, β is the thermal expansion coefficient, $\mathbf{g} = (0, -9.8)$ is the gravity acceleration, T_{ref} is a reference temperature and k the thermal diffusion coefficient.

To define the boundary conditions for problem (1) consider that

- the top and bottom boundaries are adiabatic solid walls with friction (*Hint:* set homogeneous Neumann conditions on temperature);
- the lateral boundaries are solid walls with friction on which the temperature is prescribed, namely $T = 0$ on Γ_2 and $T = 20$ on Γ_4.

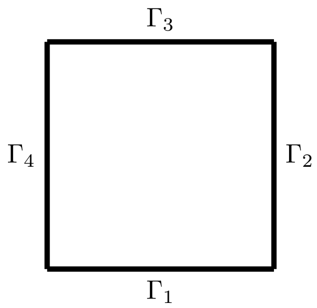

Figure 1: Computational domain

a) Define a fixed-point scheme (of Picard type) for the solution of the nonlinear problem (1) by a suitable linearization of momentum and temperature equations.

b) Write the weak formulation of the linear problems to solve at each time iteration discussing the choice of functional spaces and the treatment of boundary conditions.

c) Write a FreeFem++ script, based on the template `exam.edp`, implementing the proposed scheme based with $\mathbb{P}_2$ finite elements for the velocity and $\mathbb{P}_1$ finite elements for pressure and temperature, starting from the flow at rest and a uniform temperature $T = 10$, and chosing a suitable relative convergence criterium based on the solution increment between two iterations.

d) Compute the numerical solution for $\nu = 0.01$, $k = 0.001$, $\beta = 0.001$, $T_{\text{ref}} = 5$ and a spatial discretization size $h = 0.02$ and report the number of iterations required to reach convergence.

e) Repeat points (1)-(4) considering now the Newton method for the linearization of problem (1) and compare the results with those obtained with the fixed-point method.